Notes: Beck, Levine and Levkov (JF, 2010)

Big Bad Banks? The Winners and Losers from Bank Deregulation in the United States

1. Takeaway

Using cross-state timing of intrastate branch deregulation (1970s–1990s), the paper finds that deregulation *reduces* income inequality. Mechanically, inequality falls because incomes rise in the *lower* half of the income distribution, not because top incomes fall. Evidence is most consistent with a labor-demand channel that raises relative wages and hours of low-skilled workers.

2. Data and key variables

- Unit of observation for main results: state-year panel, $s \in \{1, \ldots, 49\}$ (48 states + DC; Delaware and South Dakota excluded), $t = 1976, \ldots, 2006$. So $49 \times 31 = 1{,}519$ observations in most state-year regressions.
- Income distribution measures (state-year), constructed from CPS (March supplement):
  - Gini coefficient (they use both log(Gini) and logistic transform).
  - Theil index (they use log(Theil) sometimes, and raw Theil when doing decompositions).
  - log(90/10): $\log(y_{90}) - \log(y_{10})$.
  - log(75/25): $\log(y_{75}) - \log(y_{25})$.
- Treatment: $D_{st} = 1$ after state s permits intrastate branching via M&As (their main deregulation date), and 0 before.
- Main controls (state-year): per-capita GSP growth, unemployment rate, share Black, share HS dropouts, share female-headed households.
- Identification intuition: timing of deregulation is driven by national tech shocks interacting with state-specific conditions; they test that pre-dereg inequality does not predict dereg timing.

3. All models/specifications in the paper

3.1 Baseline difference-in-differences (main causal model)

They estimate:

$$Y_{st} = \alpha + \beta D_{st} + \delta' X_{st} + A_s + B_t + \varepsilon_{st}, \tag{1}$$

where:

- Y_{st} is an inequality measure (logistic Gini, log Gini, log Theil, log(90/10), log(75/25)).
- A_s = state fixed effects (time-invariant differences across states).
- B_t = year fixed effects (common shocks/trends).
- X_{st} = time-varying controls (in some specs).

Interpret β: $\beta < 0$ means branch deregulation reduces inequality, controlling for national trends and fixed state traits. Standard errors are clustered at the state level.

3.2 "Exogeneity"/timing test: duration (hazard) model for deregulation

They also test whether pre-existing inequality predicts deregulation timing using a Weibull hazard/duration model (reported in Table I). In accelerated-failure-time form, a generic way to write this is:

$$\log(T_{st}) \;=\; Z'_{st}\pi \;+\; u_{st}, \qquad T_{st} = \text{time-to-deregulation,} \tag{Dur}$$

with Weibull assumptions on u_{st} (equivalently a Weibull hazard rate). They emphasize that the *Gini coefficient is not a significant predictor* of deregulation timing.

3.3 Distributional incidence: effects at each percentile of income

They compute $Y_{st}^{(i)} = \log(\text{income at percentile } i)$ for $i = 5, 10, 15, \ldots, 95$ and estimate:

$$Y_{st}^{(i)} \;=\; \alpha \;+\; \gamma D_{st} \;+\; A_s \;+\; B_t \;+\; \varepsilon_{st}. \tag{2}$$

Interpret γ: This is the average post-deregulation shift in log income at percentile i (relative to the year FE and state FE baseline). Positive significant γ at low percentiles implies the bottom rises.

3.4 Event-study dynamics for inequality (leads/lags around deregulation)

They trace dynamics by estimating:

$$\log(\text{Gini})_{st} = \alpha + \beta_1 D_{st}^{-10} + \beta_2 D_{st}^{-9} + \cdots + \beta_{25} D_{st}^{+15} + A_s + B_t + \varepsilon_{st}, \tag{3}$$

where $D_{st}^{-j} = 1$ if state s is j years *before* deregulation in year t, and $D_{st}^{+j} = 1$ if it is j years *after* deregulation; the deregulation year is omitted (so all effects are relative to year 0).
Interpret the β_j's:

- Pre-trend check: if $\beta_{-j} \approx 0$ for all $j > 0$, then inequality is not trending in advance of deregulation.
- Post effects: negative β_{+j} means inequality falls after deregulation; if it levels off, the effect looks like a level shift rather than a trend shift.

3.5 Heterogeneous effects: interaction with initial conditions

They test whether the treatment effect varies with pre-deregulation state traits that proxy for how "binding" branching restrictions were. A generic form is:

$$Y_{st} = \alpha + \beta D_{st} + \theta\big(D_{st} \times \text{InitialCond}_s\big) + A_s + B_t + \varepsilon_{st}, \tag{Het}$$

where InitialCond_s includes: unit-banking indicator, population dispersion, share of small banks, share of small firms (measured pre-dereg, often using 1976 values).
Interpret θ: If $\theta < 0$, deregulation reduces inequality more in states where restrictions were initially more distortionary.

3.6 Theil decomposition (between vs within groups)

They exploit the additivity of the Theil index to decompose inequality into between-group and within-group components. A standard decomposition is:

$$T = T^{\text{between}} + T^{\text{within}}, \tag{1}$$

$$T^{\text{between}} = \sum_g \frac{Y_g}{Y} \ln\left(\frac{Y_g/Y}{N_g/N}\right), \tag{2}$$

$$T^{\text{within}} = \sum_g \frac{Y_g}{Y} T_g, \tag{Decomp}$$

where g indexes groups (e.g., self-employed vs wage workers; or low-education vs high-education), Y_g is total income in group g, N_g is group size, and T_g is Theil within group g.
Then they regress each component on D_{st} with state and year FE to see which part changes.

3.7 Earnings inequality by age groups and conditioning on education

For earnings inequality, they create a panel with three age groups per state-year and estimate models where bank deregulation can interact with age group indicators (Table V). Conceptually:

$$\text{Ineq}_{s,t,a} = \alpha + \beta D_{st} + \beta_{36\text{–}45}\left(D_{st} \times \mathbb{1}\{a = 36\text{–}45\}\right) + \beta_{46\text{–}54}\left(D_{st} \times \mathbb{1}\{a = 46\text{–}54\}\right) + \text{FE's} + \varepsilon_{s,t,a}. \tag{Age}$$

They also compute "conditional earnings" by taking residuals from a wage regression on education dummies (plus year FE), then recompute inequality measures on those residuals, and re-run the inequality regressions. The key idea: if the effect survives conditioning on education, education attainment is unlikely to be the main mechanism.

3.8 Two-step relative wages and relative hours (labor demand channel)

Step 1: For *skilled* workers, in each year t they estimate:

$$w^s_{ist} = X_{ist}\theta^s_t + \varepsilon_{ist}, \tag{4}$$

where w^s_{ist} is log real hourly wage (skilled), and X_{ist} includes experience polynomials, gender/race indicators, and interactions; estimated separately by year to allow time-varying prices θ^s_t.

Step 2: For each *unskilled* worker, define relative wage:

$$r_{ist} = w^u_{ist} - X^u_{ist}\theta^s_t, \tag{5}$$

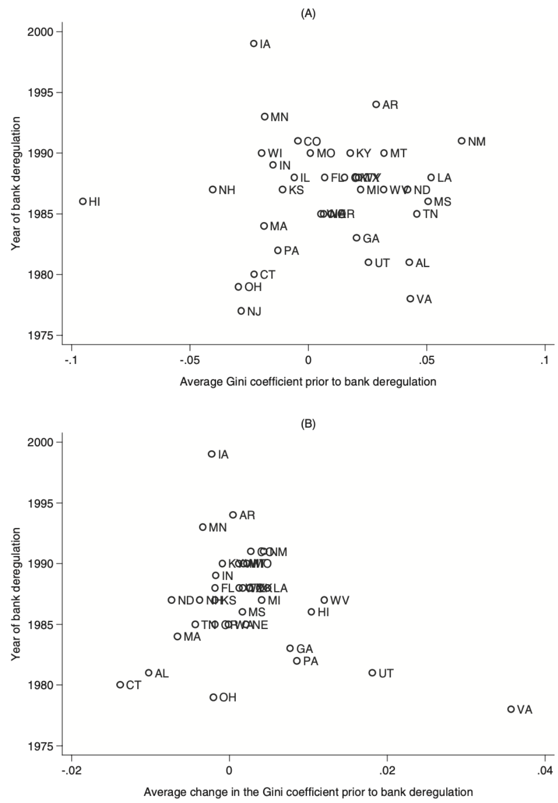

Figure 1. Timing of bank deregulation and pre-existing income inequality: Graphical

i.e., actual unskilled log wage minus the predicted skilled log wage for the same observables (including subtracting the skilled-year intercept so it is relative to the conditional mean skilled wage).

Then they run an event-study regression for relative wages (and analogously for hours):

$$r_{ist}^{(w)} = \alpha + \sum_{j=-10}^{+15} \beta_j D_{st}^{j} + A_s + B_t + \varepsilon_{ist}, \tag{6}$$

excluding year 0.

Interpret these: If deregulation raises demand for low-skilled labor, then relative wages and/or relative hours of unskilled workers should rise after deregulation.

3.9 Event-study for unemployment

They also estimate:

$$\log(\text{Unemployment})_{st} = \alpha + \sum_{j=-10}^{+15} \beta_j D_{st}^{j} + A_s + B_t + \varepsilon_{st}, \tag{7}$$

to see whether deregulation is associated with falling unemployment.

4. Figures

Figure 1: Timing of deregulation vs pre-existing inequality

- Panel (A): scatter of *average* pre-deregulation Gini vs year of deregulation.

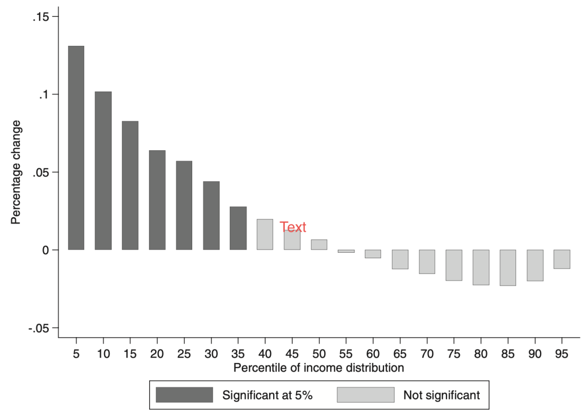

Figure 2. The impact of deregulation on different percentiles of the income distribution

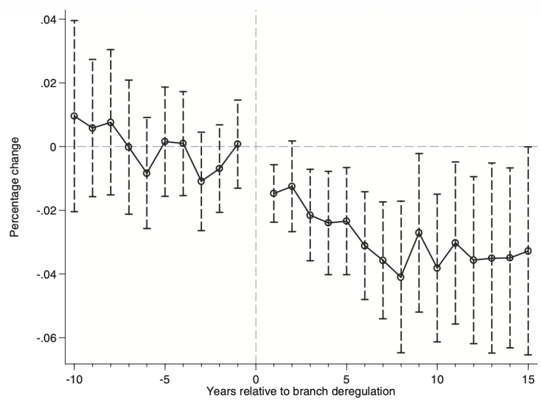

Figure 3. The dynamic impact of deregulation on the Gini coefficient of income inequality

- Panel (B): scatter of *average change* in pre-deregulation Gini vs year of deregulation.
- Reported t-stats for correlation are small (about 0.20 and -1.16), so visually/statistically there is no strong relationship.

Interpretation: This supports the identifying assumption that states did not deregulate *because* inequality was high or rising.

Figure 2: Effects on income at different percentiles

- Each bar is $\widehat{\gamma}$ from (2) for a given percentile i.
- Dark bars are significant; significant gains occur mainly below about the 40th percentile.
- No significant negative effects at high percentiles.

Interpretation: Inequality falls because the bottom rises ("pro-poor" income gains), not because the top is compressed.

Figure 3: Dynamic effect on $\log(\textbf{Gini})$ (event study)

- Pre-deregulation coefficients are near zero with no trend $\Rightarrow$ no pre-trends.
- Inequality drops quickly after deregulation; the effect grows over several years.

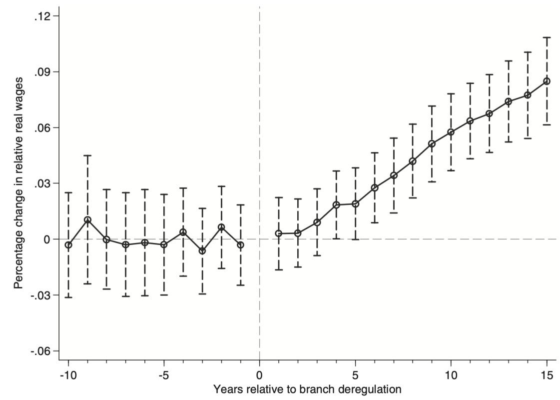

Figure 4. The impact of deregulation on the relative wages of unskilled workers. The

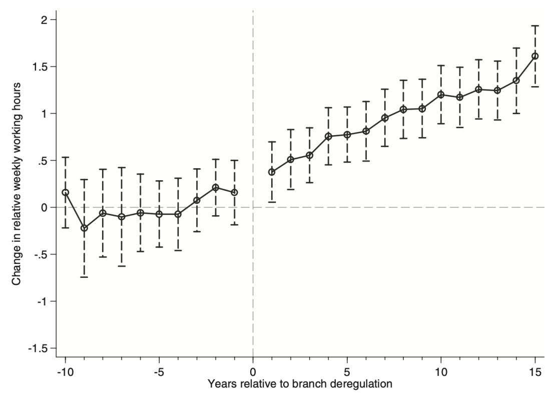

Figure 5. The impact of deregulation on the relative working hours of unskilled workers

- The effect stabilizes around an approximately 4% decline in Gini after about 8 years (level effect).

Interpretation: The mechanism operates relatively fast and looks like a permanent downward shift in inequality rather than a gradual change in trend.

Figure 4: Relative wages of unskilled vs skilled (event study)

- Relative wages increase meaningfully after deregulation (significant increase appears a few years after).
- Cumulative gain is large by year +15 (they describe it as close to 9%).

Interpretation: Consistent with increased relative demand for low-skilled labor.

Figure 5: Relative weekly hours of unskilled vs skilled (event study)

- Relative hours increase quickly post-deregulation and keep rising.
- Total effect is about +1.5 hours/week by year +15 (as described in the text).

Interpretation: Not only do low-skilled workers earn relatively higher wages, they also work relatively more hours, both of which raise their total earnings and compress inequality.

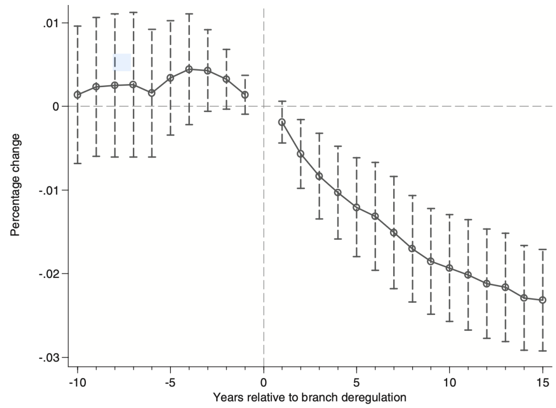

Figure 6. The impact of deregulation on unemployment rate. The figure shows the dynamic

	(1)	(2)	(3)	(4)	(5)
Gini coefficient of income inequality	0.02 (0.03)	0.02 (0.05)	0.03 (0.02)	0.03 (0.03)	0.01 (0.03)
Controls	No	Yes	No	Yes	Yes
Political-economy factors	No	No	Yes	Yes	Yes
Regional indicators	No	No	No	No	Yes
Observations	408	408	408	408	408

Figure 1: Enter Caption

Figure 6: Unemployment (event study)

- Unemployment falls starting roughly 2 years after deregulation.
- Cumulative decline is substantial by year +15 (described as more than 2 percentage points).

Interpretation: Deregulation is associated with improved labor-market conditions, reinforcing the labor-demand mechanism.

5. Tables

Interpretation of Table I.

- Across specifications, the coefficient on pre-existing inequality (Gini) is small and statistically insignificant.
- This supports the identifying assumption: inequality levels are not systematically driving deregulation timing.

Interpretation of Table II.

- The deregulation indicator is negative and statistically significant for *every* inequality measure, with and without controls.
- For log measures, read coefficients as approximate percent changes:
  - Panel A: log(Gini) coefficient $-0.022 \Rightarrow$ about a 2.2% fall in Gini ($e^{-0.022} - 1 \approx -0.022$).
  - Panel A: log(Theil) coefficient $-0.041 \Rightarrow$ about a 4.0% fall in Theil.

	Logistic Gini (1)	Log Gini (2)	Log Theil (3)	Log 90/10 (4)	Log 75/25 (5)
Panel A: No Controls					
Bank deregulation	−0.039	−0.022	−0.041	−0.135	−0.077
	(0.013)***	(0.008)***	(0.016)**	(0.058)**	(0.020)***
R^2	0.36	0.35	0.43	0.74	0.60
Observations	1,519	1,519	1,519	1,519	1,519
Panel B: With Controls					
Bank deregulation	−0.031	−0.018	−0.032	−0.101	−0.066
	(0.011)***	(0.006)***	(0.014)**	(0.050)**	(0.017)***
Growth rate of per capita GDP (2000 dollars)	−0.053	−0.028	−0.050	−0.140	−0.114
	(0.072)	(0.041)	(0.081)	(0.229)	(0.119)
Proportion blacks	−0.390	−0.218	−0.462	−0.826	−0.231
	(0.265)	(0.154)	(0.320)	(1.451)	(0.473)
Proportion high-school dropouts	0.256	0.140	0.219	0.432	−0.072
	(0.124)**	(0.071)*	(0.147)	(0.635)	(0.155)
Proportion female-headed households	0.030	0.017	0.028	0.226	0.102
	(0.100)	(0.058)	(0.125)	(0.501)	(0.153)
Unemployment rate	0.011	0.006	0.013	0.069	0.023
	(0.002)***	(0.001)***	(0.003)***	(0.014)***	(0.003)***
R^2	0.40	0.39	0.46	0.75	0.63
Observations	1,519	1,519	1,519	1,519	1,519

Table III
The Impact of Deregulation on Income Inequality as a Function of Initial State Characteristics

	(1)	(2)	(3)	(4)
Bank deregulation	−0.022	−0.026	0.026	1.797
	(0.014)	(0.013)*	(0.025)	(0.524)***
Deregulation × (unit banking)	−0.033			
	(0.017)*			
Deregulation × (initial population dispersion)		−0.313		
		(0.138)**		
Deregulation × (initial share of small banks)			−0.503	
			(0.189)**	
Deregulation × (initial share of small firms)				−2.062
				(0.590)***
Linear combination	−0.055			
	(0.017)***			
Evaluated at the 25th percentile		−0.029	−0.011	−0.011
		(0.013)**	(0.016)	(0.016)
Evaluated at the 50th percentile		−0.030	−0.029	−0.028
		(0.013)**	(0.015)*	(0.015)*
Evaluated at the 75th percentile		−0.037	−0.043	−0.046
		(0.013)***	(0.016)***	(0.015)***
Observations	1,519	1,519	1,147	1,147

  - Panel A: log(90/10) coefficient $-0.135 \Rightarrow$ about a 12.6% fall in the 90/10 ratio.
  - Panel A: log(75/25) coefficient $-0.077 \Rightarrow$ about a 7.4% fall in the 75/25 ratio.

- The result is robust to adding macro and demographic controls; unemployment is positively related to inequality.

- Bottom line: branch deregulation materially compresses the income distribution relative to national trends.

Interpretation of Table III.

- The inequality-reducing effect of deregulation is *stronger* where branching restrictions were plausibly more distortionary:
  - Unit-banking states: combined effect is -0.055 (significant), larger than in non-unit states.
  - More population dispersion: the evaluated effects become more negative as dispersion goes from 25th to 75th percentile.
  - Higher small-bank share and higher small-firm share: the evaluated effects are much more negative at the 75th percentile than at the 25th.

- This pattern fits the "bank-performance/competition" mechanism: states that were more constrained pre-reform gain more (inequality falls more).

Table IV
Decomposing the Impact of Deregulation on Income Inequality into Between- and Within-Groups

	Panel A: All Workers				
				Employment Groups	
	Total	Between Groups	Within Groups	Self Employed	Salaried
Bank deregulation	−0.0103 (0.0043)**	0.0002 (0.0003)	−0.0105 (0.0042)**	−0.0077 (0.0074)	−0.0102 (0.0042)**

	Panel B: Salaried Workers				
				Education Groups	
	Total	Between Groups	Within Groups	High School or Less	Some College or More
Bank deregulation	−0.0102 (0.0042)**	−0.0028 (0.0011)**	−0.0074 (0.0035)**	−0.0086 (0.0043)*	−0.0039 (0.0038)

Table V
The Impact of Deregulation on Earnings Inequality

	Logistic Gini (1)	Log Gini (2)	Log Theil (3)	Log 90/10 (4)	Log 75/25 (5)
Panel A: Unconditional Earnings					
Bank deregulation	−0.041 (0.018)**	−0.025 (0.011)**	−0.053 (0.023)**	−0.108 (0.038)***	−0.054 (0.020)***
(Bank deregulation) × (Ages 36–45)	0.006 (0.015)	0.004 (0.010)	0.013 (0.020)	0.034 (0.036)	−0.005 (0.015)
(Bank deregulation) × (Ages 46–54)	0.011 (0.018)	0.007 (0.012)	0.020 (0.024)	0.034 (0.042)	−0.010 (0.018)
R^2	0.26	0.26	0.26	0.40	0.39
Observations	4,557	4,557	4,557	4,557	4,557
Panel B: Earnings Conditional on Education					
Bank deregulation	−0.040 (0.015)**	−0.039 (0.015)**	−0.083 (0.031)**	−0.005 (0.002)***	−0.002 (0.001)**
(Bank deregulation) × (Ages 36–45)	0.004 (0.014)	0.004 (0.014)	0.019 (0.033)	0.001 (0.002)	−0.000 (0.001)
(Bank deregulation) × (Ages 46–54)	0.005 (0.015)	0.005 (0.015)	0.016 (0.034)	0.002 (0.002)	−0.001 (0.001)
R^2	0.50	0.50	0.40	0.48	0.46
Observations	4,557	4,557	4,557	4,557	4,557

- This also makes reverse-causality less plausible to me: the effect varies in theoretically predicted ways.

Interpretation of Table IV.

- Panel A (entrepreneurship/self-employment angle):
  - Total Theil falls (−0.0103).
  - The *between* component (self-employed vs salaried) is basically zero and insignificant.
  - Almost the entire effect comes from *within-group* inequality, especially within the salaried group.

  ⇒ Read this as: entrepreneurship or sorting into self-employment is not the main driver of the inequality decline.

- Panel B (education groups among salaried):
  - Both between-education inequality (−0.0028) and within-education inequality (−0.0074) fall.
  - The within-group fall is larger, and the reduction is stronger among the "HS or less" group.

  ⇒ The distribution tightens especially among less educated workers (consistent with a labor-demand story).

Interpretation of Table V.

- Panel A shows deregulation reduces *earnings* inequality among wage/salary workers (negative and significant across measures).
- The interaction terms with older age groups are not significant ⇒ the effect is similar across ages 25–35, 36–45, 46–54.
- Panel B conditions earnings on education (removes the part explained by education dummies), and the inequality-reducing effect remains.
- Taken together, we can interpret this as evidence against "education attainment changes are the main short-run driver" and in favor of a labor-market price/demand mechanism that raises relative pay/hours for low-skilled workers.

Conclusion, and summary

What the paper concludes

- **Main conclusion:** When U.S. states removed restrictions on *intrastate bank branching* (1970s–1990s), **income inequality fell** in those states relative to the national time trend. The paper finds a **material tightening** of the income distribution after deregulation.
- **Who gains / who loses:** Inequality fell **because the lower half of the income distribution gained**, not because high-income individuals lost. Incomes rise below about the 40th percentile; incomes above the median are largely unchanged.
- **Mechanism (their preferred channel):** The evidence points most strongly to a **labor-demand channel**: deregulation is followed by **higher relative wages and higher relative hours for unskilled workers** (relative to skilled workers), and **lower unemployment**. This combination boosts earnings at the bottom and compresses inequality.
- **Interpretation for the "big bad banks" debate:** The results do *not* support the idea that keeping banks small via branching restrictions protects the poor. Instead, branching restrictions appear to have **protected local banking monopolies** and **limited economic opportunities for lower-income groups**. Deregulation, by increasing competition and improving bank performance, disproportionately helped lower-income workers.

Summary of the paper

Research question. Inequality concerns have historically shaped banking regulation, especially restrictions on bank branching. The paper asks: **Did lifting intrastate branching restrictions make inequality better or worse?**

Empirical strategy. The authors exploit the staggered timing of intrastate branch deregulation across states and use a **difference-in-differences** design with state and year fixed effects. They also show that pre-deregulation inequality (levels and trends) does **not** predict deregulation timing (graphically and with a hazard model), supporting the plausibility of the identifying assumption.

Core results. Across multiple inequality measures (Gini, Theil, 90/10, 75/25), the estimated effect of deregulation is **negative and statistically significant**. In the event-study, inequality does **not** trend downward before deregulation, then **falls quickly afterward** and levels off: about **a 4% decline in the Gini** around eight years post-deregulation.

Incidence across the distribution. When the authors estimate the effect on log income at many percentiles, the gains are concentrated at the bottom: the **lower percentiles rise** (below roughly the 40th percentile), while there is little detectable effect above the median. Hence inequality falls because the bottom improves.

Evidence on channels. They do a set of decompositions and auxiliary outcomes:

- **Entrepreneurship channel: weak as an explanation for the inequality result.** Using a Theil decomposition between self-employed and salaried workers, essentially **none** of the inequality decline comes from narrowing the self-employed vs salaried gap. The decline is overwhelmingly **within salaried workers**.
- **Education channel: not the main driver within their sample window.** Inequality falls similarly across age groups, and the result holds when earnings are **conditioned on education** (education dummies removed). This makes it hard to attribute the main inequality drop to fast changes in educational attainment right after deregulation.
- **Labor-demand channel: strongest support.** After deregulation, **unskilled workers' relative wages rise** (eventually close to **9%** after 15 years), **their relative weekly hours rise** (about **+1.5 hours/week** after 15 years), and **unemployment falls** (more than **2 percentage points** after 15 years). These patterns line up with deregulation improving credit conditions and stimulating economic activity in a way that disproportionately increases demand for lower-skill labor.

why the effect is inequality-reducing

Step 0: what branching restrictions did. Intrastate branching limits kept banks geographically fragmented and often sheltered **local banking monopolies**. With limited competition, banks can operate less efficiently, charge higher fees/spreads, and allocate credit less effectively.

Step 1: what deregulation changes in the banking sector. Allowing branching via M&As increases contestability and consolidation, which tends to:

- increase **bank competition** (harder to extract monopoly rents),
- improve **bank efficiency** and performance,
- reduce the **cost of capital** / borrowing costs in the state economy.

Step 2: how that maps to the income distribution. Once capital gets cheaper and credit allocation improves, firms can expand and new activity is financed more easily. Even though cheaper capital can encourage substitution toward capital, the paper's evidence is that the **output/expansion effect dominates** in ways that raise **labor demand**, and in particular raise the demand for **lower-skill labor**. That shows up empirically as:

- higher relative wages for unskilled workers,
- higher relative hours worked by unskilled workers,
- lower unemployment.

When low-skill workers work more and get paid more (relative to skilled workers), incomes in the lower part of the distribution rise, and inequality falls.

Summarize the policy meaning

- The paper suggests that **regulations that limit competition among banks** can be **regressive** (disproportionately harmful to lower-income groups), at least in the historical U.S. setting of intrastate branching restrictions.
- It also suggests a broader lesson: financial development/competition can reduce inequality **primarily through labor-market improvements** (wages/hours/employment), not necessarily through more self-employment income.

What the paper is (and is not) claiming

- This is evidence about **intrastate branching deregulation** in the U.S. over 1976–2006; it is not directly a statement about all forms of deregulation or about crisis-era reforms.
- The channel analysis is presented as **preliminary** (the authors emphasize that fully pinning down mechanisms would need richer individual-level longitudinal data).
- The main results are **relative-to-trend** effects: inequality was rising nationally, but rose *less* (or fell relative to that trend) in deregulating states.